Yash Kakade

Pasadena, CA | LinkedIn | GitHub | 937-242-8610 | ykakade@caltech.edu | itsyashk.github.io

EDUCATION

California Institute of Technology **Pasadena, CA**
BS in Electrical Engineering & Computer Science GPA: 3.7/4.0 *Graduation Date: June 2027*

- Relevant Courses: Robotics Planning, Robotics Kinematics, Reinforcement Learning, Deep Learning, Computer Vision, Probability, Experimental Robotics, Robotic Systems, 3D Deep Learning, Control Systems
- Certifications: Lean Six Sigma Green Belt, Advanced Additive Manufacturing

WORK EXPERIENCE

Relativity Space **Long Beach, CA**
AI/ML Research Engineer *March 2026 – Present*

- Led weld-bead ML analysis for the first fully autonomous welding experimentation system, working alongside the Director of Autonomous Systems and company president.
- Developed Relativity Space's first physics-informed PyTorch model for weld-bead prediction, enforcing mass conservation and sparking an internal initiative to build the world's first WAAM foundation model.
- Built a weld-bead prediction benchmark adopted company-wide, implementing all 15 predictors and training the ML models for evaluation on 33,014 profiles from 29 weld runs.

California Institute of Technology **Pasadena, CA**
Robotics Machine Learning Researcher *September 2025 – March 2026*

- Built curriculum-driven Isaac Lab training environments using DFS maze synthesis and WFC constraint propagation over edge-compatible ramps, stairs, turns, and planar terrain meshes.
- Trained Unitree G1 locomotion policies with LIP-guided foot placement, velocity tracking, phase-aware contact regularization, center-of-mass stability, and torque penalties.
- Second author, *Learning Symbolic Dynamics Online for Adaptive Model-Predictive Control* (in submission).

Relativity Space **Long Beach, CA**
Robotics Software Engineering Intern *June 2025 – September 2025*

- Built a motion-capture system from scratch using Structure-from-Motion; lead inventor on the resulting patent.
- Achieved ~10 micron global accuracy in rigid-body localization, outperforming OptiTrack's published 150 micron benchmark by an order of magnitude, (IR Camera based).
- Developed a fully autonomous, self-calibrating pipeline using a custom calibration stand (no human in the loop) for faster and more robust initialization, with self-healing extrinsics based on epipolar geometry and probabilistic reprojection-error estimation.

Air Force Research Laboratory **Dayton, OH**
Robotics Software Engineering Intern *June 2024 - September 2024*

- Automated digital twins manufacturing using OptiTrack motion capture and 3D scanning for robo-simulations.
- Integrated ROS2 into the digital twin pipeline and developed a custom Dockerfile for optimized deployment and consistent environments using Kernel-Based Virtual Machines.
- Created a 3D similarity score program via Iterative Closest Point and Hausdorff distance, validate YCB dataset.
- Fine-tuned Meta SAM2 with PyTorch and YOLOv8 to create custom weights, enabling Boston Dynamics Spot to identify lab-specific objects with 92.8% accuracy.

California Institute of Technology **Pasadena, CA**
Machine Learning Researcher *July 2023 - September 2023*

- Quantified a 40% CNN accuracy drop under adversarial noise for autonomous-driving perception, then retrained on noise-affected data for a 20% accuracy gain in vehicle position identification (Python, TensorFlow).

PROJECTS (all at itsyashk.github.io/projects)

- **Rubik's Cube Robot:** 6-DOF arm that scans and solves a physical cube via ArUco/HSV vision and ROS 2.
- **Humanoid Tennis:** 29-DOF Unitree G1 striking targets with hybrid Newton–Raphson/DLS IK at 100 Hz.
- **Gita Daily Verse:** iOS App Store app (React Native, Expo, Supabase) with 100+ downloads and active users.
- **Autonomous Lawnmower:** sub-$300 ROS 2 LiDAR/SLAM mower; 2× 1st Place, Ohio State Science Fair.

AWARDS & SKILLS

Awards:

- Co-Founder & Finalist, Bill Gross Prize for Entrepreneurship, Caltech, 2026
- .406 Ventures Fellow, Class XVII, 2026; Pear Fellow
- Relativity Space Patent Inventor, accounting for 5% of the company's patent filings, 2026
- QuestBridge National College Match Scholarship Recipient 2023 (35/20800 Awarded)
- Presidential Recognition Award (Joe Biden) for Environmental Excellence for Tree Planting Drone 2022

Skills: Python, C++, Java, PyTorch, TensorFlow, CUDA, Deep Learning, CNNs, Transformers, Large Language Models, Claude Code, AI Agents, Agent Harnesses, MCP, Prompt Engineering, Reinforcement Learning, Isaac Lab, MuJoCo, Physics-Informed ML, Neural Operators, Computer Vision, OpenCV, SAM2, YOLOv8, Stereo/RGB-D Cameras, Motion Capture, ROS 2, SLAM, Docker, Linux, Git